ECO 362 Financial Investments
Final Exam

Prof. Motohiro Yogo

Fall 2023

Instructions

1. Check that you have all 5 pages. There are 6 questions.

Question	Points
1	12
2	20
3	16
4	16
5	20
6	16
Total	100

2. This is a closed book exam. However, you may use a two-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on both sides. You are under the Honor Code to comply with this requirement.

3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.

4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book:

   I pledge my honor that I have not violated the Honor Code during this examination.

5. Write your answers in the exam books. *Show all steps necessary in deriving your solution for full credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., $95.67 or 1.23%).

6. You have 3 hours to complete the exam. After 3 hours, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.

7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. Answer TRUE or FALSE. No explanation is necessary.

   [12 points: 2 points each part]

   (a) An investor with high risk tolerance should avoid long-term bonds because they have an expected return that is lower than stocks.

   (b) A passively managed mutual fund cannot earn an alpha relative to the relevant benchmark portfolio. However, the relevant benchmark portfolio may be different for different mutual funds.

   (c) An actively managed small-cap growth strategy cannot earn positive alpha relative to the Fama-French 3-factor model.

   (d) The Sharpe ratio is the most relevant metric for comparing the performance of university endowment funds, such as Princeton versus Harvard.

   (e) The amount of life insurance that one should buy is increasing in risk aversion. The more risk averse should buy more life insurance.

   (f) A pension fund has liabilities that have a longer maturity than its assets. In order to hedge itself from interest rate risk, the pension fund should enter an interest rate swap that receives fixed and pays floating.

2. According to the Gordon growth model, the price-dividend ratio of stock i is

$$\frac{P_i}{D_i} = \frac{1+G_i}{\overline{R}_i - G_i},$$

   where $\overline{R}_i$ is the expected annual return, and G_i is the expected annual dividend growth.

   Use the following table on the price-dividend ratio, the expected annual return, and the market beta for four stocks. Assume that the CAPM holds.

Stock	P/D	Expected return (%)	Beta
Bank of America	32	21.46	1.68
Coca-Cola	32	9.40	0.41
Microsoft	123		0.88
Verizon	14		0.24

   [20 points: 4 points for each part]

   (a) Which one of the four is a value stock?

   (b) Which one of the four is a growth stock?

   (c) What is the annual riskless interest rate?

   (d) What is the expected annual return for Microsoft?

   (e) What is the expected annual dividend growth for Verizon?

3. John Templeton famously invested \$100 in 1939, buying the cheapest stocks and holding them through World War II.

Use the following table on historical annual returns in percent on the overall stock market, a growth strategy, and a value strategy. The growth strategy buys stocks with the lowest book-to-market equity. The value strategy buys stocks with the highest book-to-market equity. The annual riskless interest rate is 0% from 1940 to 1945.

Year	Market	Growth	Value
1940	−7	−9	−26
1941	−10	−12	1
1942	16	14	52
1943	28	14	62
1944	21	16	69
1945	39	31	66

[16 points: 6 points for parts (a) and (b), 4 points for part (c)]

(a) Is Templeton's strategy closest to a market, growth, or value strategy? Assuming that Templeton invested \$100 in December 1939, what is your best estimate of his wealth in December 1945?

(b) Assuming that Templeton's strategy has a market beta of 1.5, what is his annual alpha relative to the CAPM?

(c) The average annual stock return is 12% from 1920 to 2021. Did Templeton earn his fortune primarily from market timing (i.e., buying stocks at the right time) or stock picking (i.e., buying the right subset of stocks)? Explain the reasoning for your answer.

4. On December 9, 2023, Shohei Ohtani signed a record deal to play for the LA Dodgers over the next 10 years. However, his salary structure is unusual. The Dodgers will pay him \$700 million over 20 years: \$2 million annually for years 1 to 10, then \$68 million annually for years 11 to 20.

   Assume that Shohei retires after 10 years with the Dodgers and has no other sources of income. His current financial wealth is \$100 million. The expected annual stock return is 12%. The standard deviation of annual stock returns is 20%.

   Use the following table on the prices of annuities per \$1 face value.

Maturity (years)	Price	Maturity (years)	Price
1	0.95	11	8.60
2	1.87	12	9.19
3	2.74	13	9.76
4	3.59	14	10.29
5	4.41	15	10.81
6	5.19	16	11.29
7	5.93	17	11.76
8	6.65	18	12.20
9	7.33	19	12.62
10	7.98	20	13.02

   [**16 points: 4 points for each part**]

   (a) What is the present discounted value of Shohei's labor income?

   (b) According to life-cycle theory, what share of Shohei's current financial wealth of \$100 million should be in stocks? Assume that his risk aversion is $\gamma = 10$.

   (c) There is speculation that the unusual salary structure may have been tax motivated. Calculate the present discounted value of Shohei's labor income <u>after taxes</u> under two scenarios. In scenario 1, Shohei lives in California for the next 20 years and faces an income tax rate of 51% (i.e., 37% for Federal plus 14% for California). In scenario 2, Shohei lives in California for the next 10 years then retires to a no-tax state like Texas, where he will pay only 37% for Federal.

   (d) Shohei's salary is guaranteed regardless of performance or injury. From Shohei's perspective, is the \$700 million contract riskless? If your answer is yes, simply state yes. If your answer is no, state no and name a source of risk that Shohei should be aware of.

5. According to the term structure model, the t-year continuously compounded zero-coupon yield is

$$y(t) = \overline{y} + \frac{1}{t}\left(\frac{1-\rho^t}{1-\rho}\right)(y(1) - \overline{y}) + \frac{t-1}{t}\mu,$$

where $\rho = 0.8$. Assume no arbitrage.

Use the following table on the continuously compounded zero-coupon yield curve.

Maturity (years)	Yield (%)
1	2.00
2	2.25
3	2.38

[20 points: 4 points for each part]

(a) Describe what $\overline{y}$ represents. Compute $\overline{y}$ based on the zero-coupon yield curve.

(b) Describe what μ represents. Compute μ based on the zero-coupon yield curve.

(c) Is the 1-year interest rate expected to rise or fall? Explain the reasoning for your answer.

(d) What is the continuously compounded yield on a consol (i.e., a zero-coupon bond with infinite maturity)?

(e) What is the continuously compounded expected annual return if you buy the 3-year zero-coupon bond and sell it after a year?

6. Nvidia stock is trading at \$466 per share. A call option on Nvidia stock with a 2-month maturity and a strike price of \$460 is trading at \$34. The continuously compounded annual riskless interest rate is 5%. Assume that Nvidia does not pay dividends. Assume no arbitrage.

[16 points: 4 points for each part]

(a) What is the futures price on Nvidia stock with a 4-month maturity?

(b) What is the price of a put option on Nvidia stock with a 2-month maturity and a strike price of \$460?

(c) Describe a portfolio strategy known as a straddle. Which assets do you need to go long or short? Draw the payoff diagram with the overall payoff on the vertical axis and the stock price on the horizontal axis. Clearly label the axes with numbers for full credit.

(d) By how much would the stock price have to move in order for you to break even (i.e., earn a payoff at least equal to the cost of the straddle strategy)? What is the straddle strategy betting on?